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MANUFACTURING OF MESAS USING DEPOSITED LAYER FOR SUBSTRATE SUPPORTS

Abstract

Embodiments of the present disclosure generally relate to forming mesas on substrate supports. In some embodiments, the substrate supports include heaters or electrostatic chucks. More specifically, embodiments relate to using physical vapor deposition (PVD) to reform damaged mesas on substrate supports by adding new layers of material. In one embodiment, a method of forming features on a substrate support is provided. The method includes depositing a first layer on a first surface of a support body of a substrate support. The method further includes forming a plurality of features in the first layer by removing portions of the first layer. The plurality of features include a plurality of valleys between a plurality of peaks. A ratio of a thickness of the support body to a thickness of the first layer is greater than 10:1.

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Background/Summary

BACKGROUND

Field

[0001] Embodiments of the present disclosure generally relate to semiconductor processing. More specifically, embodiments relate to forming mesas on substrate supports.

Description of the Related Art

[0002] Various semiconductor processing techniques implement substrate supports that may act as heaters and electrostatic chucks. These substrate supports have mesas to support the substrate during processing. Processing of the substrate can cause the mesas to wear down. Accordingly, the mesas must be repaired to continue using the substrate support in processing. Substrate supports acting as heater and electrostatic chucks have electronics such as RF meshes and electrodes disposed within the substrate support. The distance from where the substrate is placed on the substrate support and the electronics disposed within the substrate support is critical to processing performance.

[0003] Therefore, improved techniques for forming substrate mesas are needed.

SUMMARY

[0004] In one embodiment, a method of forming features on a substrate support is provided. The method includes depositing a first layer on a first surface of a support body of a substrate support. The method further includes forming a plurality of features in the first layer by removing portions of the first layer. The plurality of features include a plurality of valleys between a plurality of peaks. A ratio of a thickness of the support body to a thickness of the first layer is greater than 10:1.

[0005] In another embodiment, a method of forming mesas on a substrate support is provided. The method includes removing a first layer disposed on a first surface of a support body of a substrate support. The first layer includes a plurality of first mesas formed on the support body. The method further includes depositing a second layer on the first surface of the support body, and forming a plurality of second mesas in the second layer.

[0006] In another embodiment, a substrate support is provided. The substrate support includes a support body having a first surface, and a plurality of mesas formed in a first layer. The plurality of mesas are disposed on the first surface of the support body. A ratio of a thickness of the support body to a thickness of the first layer is greater than 10:1. The substrate support further includes a RF mesh disposed in the support body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, may admit to other equally effective embodiments.

[0008] FIG. 1 is a schematic, side view of a processing chamber with a substrate support, according to embodiments.

[0009] FIG. 2A is a schematic, side view of a support body of a substrate support, according to embodiments.

[0010] FIG. 2B is a schematic, side view of a support body of a substrate support, according to embodiments.

[0011] FIG. 2C is a schematic, side view of a support body of a substrate support, according to embodiments.

[0012] FIG. 3 is a flow diagram of a method for repairing a substrate support, according to embodiments.

[0013] FIGS. 4A-4D are schematic, cross-sectional views of a substrate support during a method, according to the embodiments

[0014] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0015] Embodiments of the present disclosure generally relate to forming mesas on substrate supports. In some embodiments, the substrate supports include heaters or electrostatic chucks. More specifically, embodiments relate to using physical vapor deposition (PVD) to reform damaged mesas on substrate supports by adding new layers of material.

[0016] Mesas are present on substrate supports to support a substrate without the substrate lying flat in complete contact with the substrate support. Mesas can wear down with repeated use in process chambers. Because substrate supports are expensive, repair is preferable to replacement. Normally, the mesas are formed on the substrate support at the time of fabrication of the substrate support. The method of repair is to resurface the mesas by grinding and reforming the mesas. The damaged mesas are grinded away, and new mesas are formed from bulk material of the substrate support. Grinding the mesas reduces buffer material between a top surface of the mesas and a RF mesh disposed in the substrate support. The buffer material prevents leakage of voltage and prevents drifting of the electrical charge during chucking. The buffer material limits the amount of times the mesas may be replaced in the life cycle of the substrate support.

[0017] Various techniques described herein allow for repeated replacement of mesas during the life cycle of the substrate support. The mesas are partially or fully formed from a physical vapor deposition (PVD) material layer. The mesas made of PVD material may be removed and replaced without affecting the amount of buffer material between the RF mesh and the mesas. The PVD material layer may provide additional properties to enhance the substrate support. By having near unlimited mesa replacement, the cost of ownership of the substrate support is significantly reduced. The use of PVD to form the mesas allows for optimization to reduce current drift over prolonged use of chucking prior to stabilization.

[0018] FIG. 1 illustrates a schematic view of a process chamber **100** according to some embodiments of the disclosure. The process chamber **100** includes a chamber body **102** and a lid **104** defining a process volume **114** therein. A bottom **124** of the chamber body **102** is opposite the lid **104**. A port **106** is formed through the lid **104**. A gas source **108** is in fluid communication with the port **106**. A showerhead **110** is coupled to the lid **104**. A plurality of openings **112** are formed through the showerhead **110**. The gas source **108** is in fluid communication with the process volume **114** via the port **106** and the openings **112**.

[0019] A substrate support **116** is moveably disposed in the process volume **114** opposite the lid **104**. The substrate support **116** includes a support body **200** disposed on a stem **118**. The support body **200** includes a support surface **201** disposed opposite the stem **118** and facing the showerhead **110**. In some embodiments, the support body **200** includes a heater or an electrostatic chuck. The support body **200** is formed from a bulk material. The support body **200** may include aluminum nitride (AlN), boron nitride (BN), or a similar material. The support body **200** may have a thickness **202** ranging from 15 mm to 75 mm, such as 25 mm to 52 mm. The support surface **201** includes a plurality of mesas **203**, which are further described in FIGS. 2A and 2B. Although the techniques disclosed herein are described with respect to forming mesas **203** on a substrate support **116**, any of the techniques described herein may be implemented to form any type of feature on a substrate

support **116**, including mesas, ledges, pockets, or tabs. An opening **120** is formed through the chamber body **102** between the lid **104** and the bottom **124**. During operation, a substrate **101** is loaded onto the support surface **201** through the opening **120**. An actuator **126** is coupled to the substrate support **116** to move the substrate support **116** toward and away from the showerhead **110** for loading and processing the substrate **101** thereon.

[0020] An RF mesh **122** is disposed within the support body **200**. One or more portions of the RF mesh **122** are disposed in a plane that is substantially perpendicular to the support surface **201**. In some embodiments, a heating element is disposed within the support body **200**. The heating element and RF mesh **122** may be used to heat the substrate **101** or electrostatically chuck the substrate **101**. The heating element may be disposed below the RF mesh **122** and may be substantially perpendicular to the support surface **201**. The RF mesh **122** is a set distance away from the support surface **201** described further in FIGS. 2A and 2B. The RF mesh **122** is connected to one or more RF leads **127**. The RF leads **127** are coupled to an RF power source **128**. The RF power source **128** provides RF power to the RF mesh **122**. Although the processing chamber **100** described herein is provided as an example of a chamber with which the substrate support **116** can be implemented, in various embodiments, the substrate support **116** can be implemented with other types of processing chambers.

[0021] FIG. 2A is a schematic, side view of a first support body **200A**. The first support body **200A** is shown in the process chamber **100** in FIG. 1. As described above, the support body **200A** includes the support surface **201**, the mesas **203**, and the RF mesh **122**. The support body **200A** further includes an edge ring **205**. In some embodiments, the edge ring **205** is formed from the support body **200A** and extends around the mesas **203**. The edge ring **205** also surrounds the substrate **101** when the substrate **101** is placed on the mesas **203**. The substrate **101** is placed on the support surface **201** of the mesas **203**. The support surface **201** is part of the mesas **203**.

[0022] Each of the one or more mesas **203** is formed between two or more valleys **207**. Each of the one or more valleys **207** may be formed by removing material from the support surface **201** between each of the one or more mesas **203**. The mesas **203** may be a dimple pattern. The mesas **203** may be an isolated raised feature. The feature may be rectangular, square, round, etc. The pattern can have any number of mesas **203** at any coordinates within the support surface **201**. The mesas **203** prevent the substrate from contacting a first surface **209** of the support body **200A**.

[0023] In FIG. 2A, the mesas **203** are formed from a deposited layer **211**. In some embodiments, the deposited layer **211** may be deposited by physical vapor deposition (PVD). Other methods for depositing the deposited layer **211** include, for example, atomic layer deposition (ALD), thermal spray (e.g. suspension plasma spray, vacuum plasma spray, or similar processes), sol-gel, and other dense, thick film deposition techniques. The deposited layer **211** may include at least one of aluminum nitride (AlN), Aluminum oxynitride (Al.sub.xO.sub.yN.sub.z), low vapor pressure compounds or a similar ceramic material. Low vapor pressure compounds include single metal oxides (M-O), single metal oxyfluorides (M-O-F), single metal fluorides (M-F), double metal oxides (M1M2O), double metal oxyfluorides (M1M2OF), double metal fluorides (M1M2F), triple metal oxides (M1M2M3O), triple metal oxyfluorides (M1M2M3OF), and single metal fluorides (M1M2M3F). The metals (e.g., M, M1, M2, M3) used in the low vapor pressure compounds may include, for example, magnesium (Mg), erbium (Er), strontium (Sr), calcium (Ca), yttrium (Y), barium (Ba), lanthanum (La), samarium (Sm), and similar metals. The material of the deposited layer **211** provides benefits to the mesas **203** such as thermal, mechanical, and physical property improvements compared to the bulk material. In some embodiments, the deposited layer **211** includes multiple layers that form multi-compositional mesas **203**.

[0024] In some embodiments, the deposited layer **211** is made of the same material as the bulk material of the support body **200A**. In other embodiments, the deposited layer **211** includes a material different to the bulk material of the support body **200A**. In some embodiments, the deposited layer **211** includes multiple layers of different materials, for example, in order to reduce a

mismatch of the coefficients of thermal expansion (CTEs) between the support body **200** and the mesas **203**. For example, in some embodiments, if the support body **200** and the mesas **203** have different CTEs, one or more layers having CTE(s) that are in between the CTE of the support body **200** and the CTE of the mesas **203** may be disposed between the support body **200** and the mesas **203** in order to more gradually transition between the different CTEs. In some embodiments, the deposited layer **211** has multiple layers tuned for intended processing temperature. The deposited layer **211** may be a material to give the mesas **203** additional properties, which may be tailored to processes performed in the process chamber **100**, for example, reducing leakage current from the RF mesh **122** or reduce backside substrate scratching. In some embodiments, the deposited layer **211** is partially or fully fluorinated to avoid further build-up of fluoride during processing. The deposition of the deposited layer **211** is described in FIG. 3. The deposited layer **211** forms both the peaks **208** and the valleys **207** of the plurality of mesas **203**. The deposited layer **211** covers both the peaks **208** and valleys **207**. The first surface **209** is covered by the deposited layer **211** across the plurality of mesas **203**. The bulk material is covered by the deposited layer **211** across the plurality of mesas **203**. The first surface **209** is flat. Benefits of the support body **200A** include design freedom to change the pattern of the mesas **203** every repair cycle.

[0025] Creating the valleys **207** forms peaks **208** on each of the mesas **203**. The peaks **208** form the support surface **201**. The valleys **207** are formed by removing portions of the deposited layer **211**. The bottom of the valleys **207** includes a portion of the deposited layer **211** such that a portion of the support body **200** is not removed to form the mesas **203**. The support body **200** is completely protected by the deposited layer **211** across the mesas **203**. The deposited layer **211** has a layer thickness **204**. The layer thickness **204** is governed by a height of the mesas **203**. The height of the mesas **203** is defined by a distance between the valleys **207** and the peaks **208**. In various embodiments, the height of the mesas is greater than about 10% of the layer thickness **204**, such as about 90% to 10%, such as 75% to 15%, such as 70% to 35%. The layer thickness **204** ranges from 1 micron to 2500 microns, such as 10 microns to 1000 microns, such as 20 microns to 500 microns. In some embodiments, a ratio of the thickness **202** of the support body **200** to the layer thickness **204** is greater than 10:1, such as ranging from about 20:1 to about 50,000:1, such as about 100:1 to 5,000:1.

[0026] FIG. 2B is a schematic, side view of a second support body **200B**. The second support body **200B** includes the support surface **201**, the mesas **203**, the RF mesh **122**, and the edge ring **205**. In FIG. 2B, the mesas **203** are formed from the deposited layer **211** and the support body **200B**. The first surface **209** follows a pattern formed by the second portion **215** of the mesas **203**. The deposited layer **211** form a portion of the mesas **203**. A first portion **213** of the mesas **203** is formed from the deposited layer **211**. The deposited layer **211** may be one of the deposited layers **211** described above. The deposited layer **211** may include any of the materials and properties described above. A second portion **215** of the mesas **203** is formed from the support body **200B**. The first surface **209** of the support body **200B** is a top surface of the second portion **215** and contacts the first portion **213**. The patterns of the first portion **213** and the second portion **215** substantially match one another. The support surface **201** is a top surface of the first portion **213** of the mesas **203**. In some embodiments, the first portion **213** and second portion **215** have the same thickness. In other embodiments, the first portion **213** and second portion **215** have different thicknesses. In some embodiments, the first portion **213** has a constant thickness. Since the support body **200B** forms a portion of the mesas **203**, the mesas design cannot be changed during reforming. The support body **200B** has the benefit of shorter repair time and requiring less material to form new mesas **203**. The mesas **203** are formed in the same way as described above, except the support body **200B** forms a second portion **215** of the mesas **203**. The deposited layer **211** covers the plurality of the second portion **215** such that the support body **200B** is not removed to form the mesas **203**. The support body **200B** is completely protected by the deposited layer **211** across the mesas **203**.

[0027] As described above, the RF mesh **122** is disposed in the support body **200**. In order to

function properly, the RF mesh **122** is a set distance **217** from the support surface **201**. The distance from the support surface **201** allows material from both the mesas **203** and support body **200** to separate the RF mesh **122** and the substrate **101** to be placed on the support surface **201**. The distance causes the beneficial properties described above. The set distance **217** varies between substrates, processes, and substrate supports. The set distance **217** is about 0.5 mm to about 2.5 mm, such as 1 mm. As the substrate support **116** is used in the process chamber **100**, the mesas **203** are worn down. The mesas **203** may be replaced when the mesas **203** negatively affect processing performance. Examples of issues caused by defective mesas **203** include backside damage to the substrate **101** such as backside scratches, loss of deposition uniformity on the substrate **101**, and other similar issues.

[0028] FIG. 2C is a schematic, side view of the support body **200A**, according to embodiments. FIG. 2C illustrates embodiments in which the edge ring **205** is formed from the deposited layer **211** and extends around the mesas **203**. FIGS. 2A and 2B illustrate embodiments in which the edge ring **205** is formed in the support body **200**. The layer thickness **204** of the deposited layer **211** includes the edge ring **205** when the edge ring **205** is formed from the deposited layer **211**. When the mesas **203** are reformed as described in FIG. 3, the edge ring **205** may be reformed as well.

[0029] FIG. 3 is a flow diagram describing steps of a method **300** of reforming mesas **203**. FIGS. 4A-4D illustrate the substrate support **116** during the method **300**. Although the techniques are described with respect to forming mesas **203** on a substrate support **116**, the techniques described herein may be implemented to form any type of feature on a substrate support **116**, including mesas, ledges, pockets, or tabs

[0030] Prior to operation **301**, FIG. 4A illustrates the substrate support **116** having first mesas **403A** that have been worn down, for example, during semiconductor processing. The first mesas **403A** need to be reformed. In various embodiments, the first mesas **403A** can no longer support the substrate **101** during processing. Examples of issues caused by defective first mesas **403A** include backside damage to the substrate **101**, such as backside scratches, loss of deposition uniformity on the substrate **101**, and other similar issues. In some embodiments, the determination of whether the first mesas **403A** need to be reformed is an automated process. A controller connected to the process chamber **100** has sensors to analyze the first mesas **403A**. The controller makes a determination using the factors above to determine when the first mesas **403A** need to be reformed. In some embodiments, as shown in FIGS. 2A and 4A, the first mesas **403A** are formed entirely out of a first layer **411A** which corresponds to the deposited layer **211**. In other embodiments, such as shown in FIG. 2B, the first mesas **403A** are partially formed out of the deposited layer **211** and partially formed out of the support body **200B**.

[0031] At operation **301**, the first layer **411A** is removed. In some embodiments, the first layer is formed by PVD, atomic layer deposition (ALD), thermal spray (e.g. suspension plasma spray, vacuum plasma spray, or similar processes), sol-gel, and other dense, thick film deposition techniques. The first layer **411A** is removed using at least one of a mechanical process, like machining, bead blasting, laser patterning, etc., and a chemical process, like lithography or etching. FIG. 4B shows the substrate support **116** with the first layer **411A** removed. As shown in FIG. 4B, the first mesas **403A** are entirely formed from the first layer **411A**. The first surface **209** of the support body **200A** is exposed. The first surface **209** is flat. In embodiments with the support body **200B**, the first mesas **403A** are partially formed with the first layer **411A** and partially formed with the bulk material of the support body **200B**. The first surface **209** of the support body **200B** is exposed. The first surface **209** forms the second portion **215** of the first mesas **403A**. The second portion **215** remains while the first portion **213** is removed.

[0032] At operation **303**, a second layer **411B** is deposited. The second layer **411B** is deposited using PVD, atomic layer deposition (ALD), thermal spray (e.g. suspension plasma spray, vacuum plasma spray, or similar processes), sol-gel, and other dense, thick film deposition techniques. The second layer **411B** also corresponds to the deposited layer **211**. In some embodiments, second layer

411B includes multiple layers. FIG. 4C shows the substrate support **116** with the second layer **411B** deposited. FIG. 4C shows a uniform thickness second layer **411B** disposed on the support body **200A**. The second mesas **403B** are to be formed from the second layer **411B**. In embodiments that implement the support body **200B**, the second layer **411B** is disposed on the pattern formed by the second portion **215** to form the second mesas **403B**. In some embodiments, the substrate support **116** is first formed, and operation **301** is skipped, since no first layer **411A** is present on the support body **200**. In such embodiments, at operation **303**, a layer of material is deposited on the support body **200A** for the first time. In some embodiments, a layer of material is deposited on the support body **200B** for the first time. Performing operation **303** on the original manufactured substrate support **116** allows the method **300** to be performed to refurbish the substrate support **116** later. [0033] At operation **305**, the second mesas **403B** are formed in the second layer **411B**. In some embodiments, the mesas **203** are formed using bead blasting, laser, photolithography, or other machining techniques. Bead blasting uses a pattern transfer to pattern the second mesas **403B**. The pattern transfer uses a mask to block beads from removing material in sections with the mesa **203** and allow beads to remove material to form valleys **207** and peaks **208**. The valleys **207** are formed by removing portions of the second layer **411B**. The bottom of the valleys **207** includes a portion of the second layer **411B** such that the support body **200** is not removed to form the second mesas **403B**. In embodiments where the edge ring **205** is formed from the first layer **411A**, the edge ring **205** may be reformed from the second layer **411B**. FIG. 4D shows the substrate support **116** with the second mesas **403B** formed. The second mesas **403B** have the support surface **201**, which supports the substrate **101**. The support surface **201** is the set distance **217** away from the RF mesh **122**. In some embodiments with the support body **200B**, the pattern of the second portion **215** is the same as the pattern of the second mesas **403B**. The second layer **411B** then does not need to be machined as the uniform thickness completes the second mesas **403B**. In other embodiments, the second layer **411B** disposed on the support body **200B** is machined as described above.

[0034] Multiple substrate supports **116** can be repaired at the same time using the method **300**. For example, multiple substrate supports **116** can be placed into a batch chamber to have the method **300** performed. Performing the method **300** on multiple substrate supports **116** at the same time further reduces the time and cost of refurbishing the mesas **203**.

[0035] In summation, embodiments of the present disclosure generally relate to mesas on substrate supports. The substrate supports can include heaters or electrostatic chucks. More specifically, embodiments relate to using physical vapor deposition to reform damaged mesas on substrate supports. The method allows for near unlimited mesa replacement in the life cycle of the substrate support. The mesas are partially or fully formed from a physical vapor deposition (PVD) material layer. Benefits include the mesas being removed and replaced without affecting the amount of material between the RF mesh and mesa, and endowing additional properties to enhance the substrate support provided by the PVD material. The cost of ownership of the substrate support is significantly reduced due to the mesas being replaceable without affecting the rest of the substrate support. The use of PVD to form the mesas allows for optimization to reduce current drift over prolonged use of chucking prior to stabilization. Ease of replacing mesas will reduce backside scratching of substrates during processing.

[0036] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A method of forming features on a substrate support, comprising: depositing a first layer on a first surface of a support body of a substrate support; and forming a plurality of features in the first layer by removing portions of the first layer, wherein the plurality of features comprise a plurality

of valleys between a plurality of peaks, wherein a ratio of a thickness of the support body to a thickness of the first layer is greater than 10:1.

2. The method of claim 1, wherein a height of the plurality of features is greater than about 10% of a thickness of the first layer.
 3. The method of claim 1, wherein the first layer comprises at least one of aluminum nitride, or aluminum oxynitride, and the first layer is deposited by physical vapor deposition (PVD), thermal spray, sol-gel, or atomic layer deposition (ALD).
 4. The method of claim 1, wherein the support body comprises aluminum nitride or boron nitride.
 5. The method of claim 1, wherein an edge ring is formed from the first layer along with the plurality of features, the edge ring disposed on the support body and surrounding the plurality of features.
 6. The method of claim 1, wherein the first surface of the support body includes a pattern that corresponds to the plurality of features.
 7. A method of forming mesas on a substrate support, comprising: removing a first layer disposed on a first surface of a support body of a substrate support, the first layer including a plurality of first mesas formed on the support body; depositing a second layer on the first surface of the support body; and forming a plurality of second mesas in the second layer.
 8. The method of claim 7, wherein a ratio of a thickness of the support body to a thickness of the second layer is greater than 10:1.
 9. The method of claim 7, wherein a height of the plurality of second mesas is greater than about 10% of a thickness of the second layer.
 10. The method of claim 7, wherein the first layer and the second layer comprise at least one of aluminum nitride or aluminum oxynitride, and the first layer and the second layer are deposited by physical vapor deposition (PVD).
 11. The method of claim 7, wherein the support body comprises aluminum nitride.
 12. The method of claim 7, wherein an edge ring is formed from the second layer along with the plurality of second mesas, the edge ring disposed on the support body and surrounding the plurality of second mesas.
 13. The method of claim 7, wherein the first surface of the support body includes a pattern that corresponds to the plurality of second mesas.
 14. A substrate support, comprising: a support body having a first surface; a plurality of mesas formed in a first layer disposed on the first surface of the support body, wherein a ratio of a thickness of the support body to a thickness of the first layer is greater than 10:1; and a RF mesh disposed in the support body.
 15. The substrate support of claim 14, wherein a height of the plurality of mesas is greater than about 10% of a thickness of the first layer.
 16. The substrate support of claim 14, wherein the support body comprises at least one of a heater or an electrostatic chuck.
 17. The substrate support of claim 14, wherein the first layer comprises at least one of aluminum nitride or aluminum oxynitride, and the first layer is deposited by physical vapor deposition (PVD).
 18. The substrate support of claim 14, further comprising an edge ring formed from the first layer, the edge ring disposed on the support body and surrounding the plurality of mesas.
 19. The substrate support of claim 14, wherein the plurality of mesas comprise: a first portion comprising the first layer; and a second portion formed on the first surface of the support body, wherein the first portion is disposed on the second portion.
 20. The substrate support of claim 14, further comprising one or more layers disposed between the support body and the first layer, wherein the one or more layers have a coefficient of thermal expansion (CTE) that is between a first CTE of the support body and a second CTE of the first layer.
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